Transfer learning, Finetuning -

Transfer learning is process of taking knowledge learned from one task and applying it to a different but related task.

In the context of large language models (LLMs), it consists of two main phases.

1. Pre training - Learning general patterns from large scale unlabelled data.
2. Fine tuning - Adapting the pretrained model to specific downstream tasks.

Training LLM from scratch is expensive and it needs large amounts of data and large amounts of compute.

Hence we take a pre-trained model and fine tune it for specific task.

3 stages of LLM training pipeline :-

Stage 1. Pretraining (Language modelling)

- Create foundational knowledge of language.
- Process :- trains on massive unlabelled text corpora.

  Predict next token given previous context.

  given : $x_1, x_2, \ldots x_T$ model learns to maximise

$$\mathcal{L}_{LM} = \sum_{t=1}^{T} \log P(x_t | x_1, x_2, \ldots, x_{t-1}; \theta)$$

  where θ represents model parameters

The model learns :

- Grammar and syntax
- World knowledge
- Semantic relationships
- Contextual understanding

Stage 2 : Supervised fine tuning (SFT)

Train model to follow instructions and perform specific tasks.

Base models are only trained for next token predictions, not instruction following. Hence we train model on instruction response pairs.

Model learns to interpret user intent.

Uses smaller dataset than pretraining

Stage 3: Preference tuning

To further personalise model according to usecase or user preferences:-

we use:- 1) RLHF

2) DPO (Direct preference optimisation)

3) constitutional AI (predefined restrictions using principles and values baken into LLM.

- Improves response quality
- Reduces harmful o/p
- Aligns with specific usecases
- Makes model even more task specific
- Reduces bias, better tone and style.

Parameter efficient fine Tuning (PEFT)

Instead of updating all parameters update only a small subset while keeping pre-trained weights frozen.

Goal: Achieve comparable performance to full fine-tuning with < 5% trainable parameters.

Advantages over full fine tuning:-

- Less cost of training
- Faster training
- Low storage requirements.

Main approaches of PEFT:-

Adapter layers: Insert small bottleneck modules.

Prompt tuning: Learn task specific soft prompts

Prefix tuning: Prepend trainable prefix vectors

LORA: Low-rank weight update matrices

$(IA)^3$: Learned element wise rescalling

Adapters:-

Adapters are small, trainable neural network modules inserted between existing layers of a pre-trained model.

- Tuned to improve performance on specific tasks.
- original pre-trained weights remain frozen
- only adapter parameters are updated during fine-tuning.